

Final Capstone Project

1. Overview:

This report presents an analysis of factors influencing the success of SpaceX Falcon 9 first stage landings. Understanding the critical factors that influence successful landings is essential for improving future missions and cutting expenses as SpaceX pushes the limits of reusable rocket technology. The impact of the launch site, trends in landing success over time, and the effectiveness of different machine learning models in forecasting landing outcomes are all examined in this study using publicly available data on SpaceX launches.

2. Data Collection

The SpaceX REST API, which offers comprehensive details on individual launches, served as the main

source of data for this analysis. Though the provided code snippet for Wikipedia extraction is a placeholder and would require additional development to fully integrate historical data, the intention was to extract additional data, especially historical launch information, from Wikipedia pages.

3. Data Wrangling:

To extract pertinent features for the analysis, the original dataset from the SpaceX API was processed. Important actions included:

- **Feature Selection:** Finding and choosing columns like "fairings," "links," "net," "date_utc," "success," "cores," "capsules," "payloads," "launchpad," "flight_number," and "name" that are relevant to landing success.
- **Landing Outcome Extraction:** Creating a function to ascertain the landing success based

on data nested within the "cores" and "fairings" columns is known as "landing outcome extraction."

- **Managing Missing Values:** Resolving missing values in important columns such as "success," "date_utc," "launchpad," and the recently added "landing_success" column. Rows lacking critical information were eliminated, and missing 'landing_success' values were imputed as False. False was also used to fill in the missing "net" values.
- **Feature Engineering:** involves extracting the "year" from "date_utc" and converting it to datetime objects. Additionally, the 'launchpad' ID was extracted.
- **One-Hot Encoding:** To get the categorical 'launchpad_id' column ready for machine learning models, one-hot encoding is applied.

- Data Cleaning: Removing the date/launchpad and nested columns.
- Preparing the Target Variable: Making sure the "landing_success" target variable is formatted appropriately (integer 0 or 1) for classification models.

4. Exploratory Data Analysis:

The exploratory data analysis revealed several key insights into the factors influencing landing success:

- Trend Over Time: The landing success rate has shown a distinct and noteworthy upward trend over time, demonstrating SpaceX's ongoing technological advancements and operational expertise. Additionally, the launch success rate is trending upward and has recently stabilized at 100%.

- Launchpad Influence: The average landing success rate seems to be significantly influenced by the launchpad that is utilized for a mission. Some launchpads demonstrate higher success rates than others.
- Launch Success Correlation: First stage landing success has a strong correlation with overall mission success. A successful landing attempt is much more likely to follow a successful launch.
- Net Success Correlation: In a similar vein, landing success and "net" success (which pertains to fairing recovery) have a positive correlation, despite the fact that the "net" column itself contained missing values that were filled in by imputation.
- Correlation Heatmap: The correlation heatmap illustrates the impact of particular launchpads (shown by their one-hot encoded columns) on landing success and graphically validates the

positive correlation between "year," "success," and "landing_success."

5. Predictive Analysis:

Based on the prepared features, a number of classification algorithms were used to forecast the success of first stage landings:

Model Evaluation: The model that were accessed are K-Nearest Neighbors (KNN), Decision Trees, Support Vector Machines (SVM), and Logistic Regression.

Feature Set: 'year','success', 'net', and the one-hot encoded 'launchpad_id' columns were used as features to train the models.

Methodology: Training and testing sets of data were separated. Using GridSearchCV with cross-validation on the training set, basic hyperparameter tuning was carried out. A variety of metrics were

then used to assess the adjusted models on the test set that was not visible.

6. Results:

The results obtained are as follows:

Logistic Regression: With an accuracy of 0.8684 and a perfect recall of 1.0000 on the test set, logistic regression demonstrated its capacity to accurately identify every successful landing.

Support Vector Machine: With an accuracy of 0.9211, a precision of 0.9688, and an F1-score of 0.9538 on the test set, the Support Vector Machine (SVM) demonstrated excellent performance.

Decision Tree: On the test set, the decision tree's accuracy was 0.8158.

K-Nearest Neighbors: With an accuracy of 0.9211, a precision of 0.9688, and an F1-score of 0.9538

on the test set, K-Nearest Neighbors (KNN) performed similarly to SVM.

7. Conclusion:

This analysis shows that a number of factors, including the launchpad used, the year of the launch (which reflects experience and technological advancements), and the launch mission's overall success, affect the success of SpaceX Falcon 9 first stage landings. It is clear that landing success has been rising over time, underscoring SpaceX's advancements in reusable rocket technology. Based on the available features, predictive modeling efforts show that machine learning algorithms like SVM and KNN can accurately predict landing outcomes. This could be useful for risk assessment and future mission planning.